

Apache Maven - Installation & Setup

Apache Maven: concept, Quality Engineering use cases, first steps and official reference.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

<https://maven.apache.org/download.cgi>

<https://maven.apache.org/guides/>

Step-by-step installation and setup

1. Follow this tool-specific step: Acesse maven.apache.org e baixe o binary archive.
2. Follow this tool-specific step: Extraia para uma pasta estável, por exemplo `C:\Tools\apache-maven-<versão>`.
3. Follow this tool-specific step: Adicione a pasta bin do Maven ao Path. MAVEN_HOME pode ser configurado quando sua organização/ferramenta exigir.
4. Follow this tool-specific step: Abra novo terminal e execute `mvn -version`.
5. Follow this tool-specific step: Confirme que a saída mostra Maven e o Java esperado.
6. Follow this tool-specific step: Crie um projeto de exemplo com o archetype oficial ou use um pom.xml existente e execute `mvn test`.

Commands and validation

```
mvn -version
```

```
mvn test
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes



mvn não reconhecido: revise Path para apache-maven/bin.

Java home incorreto: corrija JAVA_HOME e valide java -version.

Falha ao baixar dependências: investigue proxy/repositório/certificado antes de alterar o projeto.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.